

1. **True or false: correct any false statements**
2. 1. Alpha particles are the most penetrating
3. 2. Alpha particles are the most ionising
4. 3. Gamma rays are able to penetrate paper
5. 4. Beta particles are 2 protons and 2 neutrons
6. 5. Beta particles can penetrate paper
7. 6. In beta decay a proton changes into a neutron and an electron
8. 7. Gamma decay changes the nucleus
9. 8. Alpha decay changes the nucleus

Answer the following questions in your book:

1. Write the nuclear equation when $^{215}_{84}\text{Po}$ undergoes alpha decay
2. Write the nuclear equation when $^{215}_{84}\text{Po}$ undergoes beta decay
3. State what is happening when a nucleus undergoes beta decay

Radioactive half-life

Learning Objective

- Explain what is meant by radioactive half-life
- Calculate the half-life
- Calculate the decline in activity after a number of half-lives

Half-life:

- Can we predict when a nucleus will decay?
- 1 count is 1 decay
- A nucleus may start off very active but if it has a high activity it will stop being radioactive quickly

Activity:

The activity is defined as the rate at which the unstable nuclei from a source of radiation decays

Activity is measured in Becquerels (Bq)

1 Becquerel is equal to 1 nucleus in the source decaying in 1 second

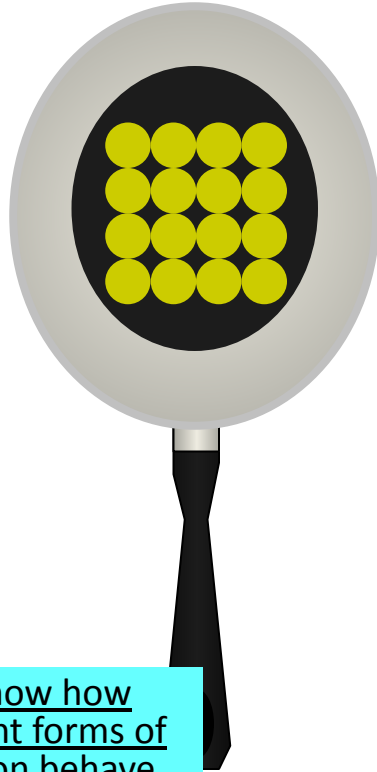
What is “Half Life”?

The time taken for
half the radioactive
atoms to decay

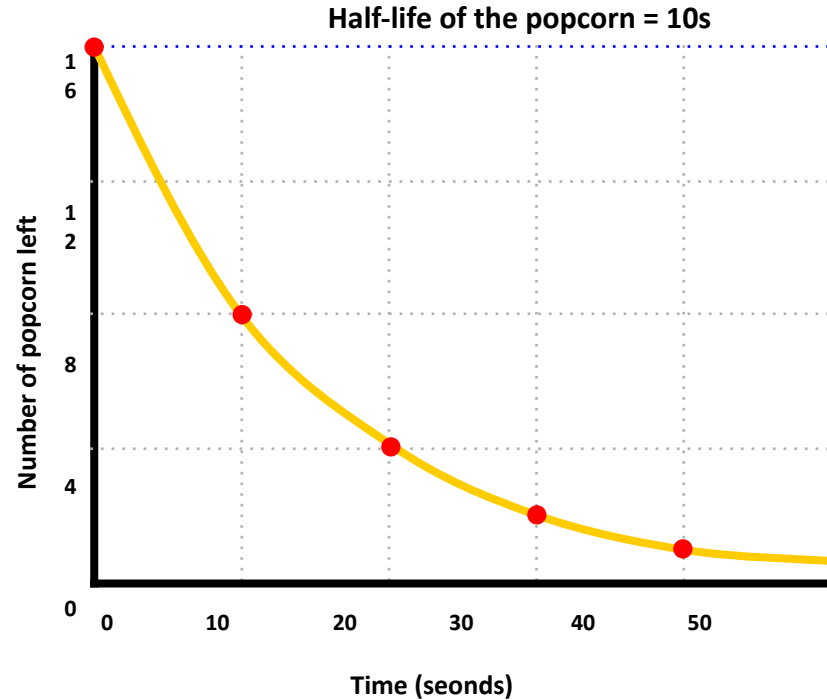
OR

The time
radioactivity takes
to reach half its
dose

Example of half-life, using a pan of popping popcorn



To know how
different forms of
radiation behave.

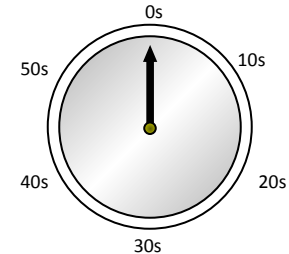
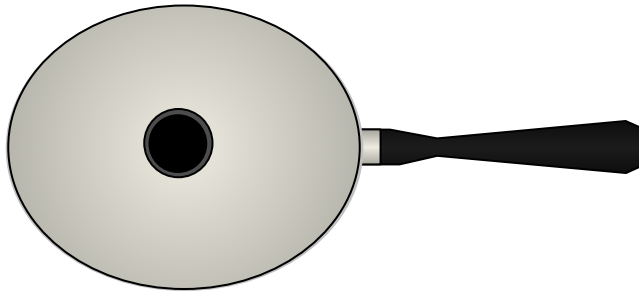
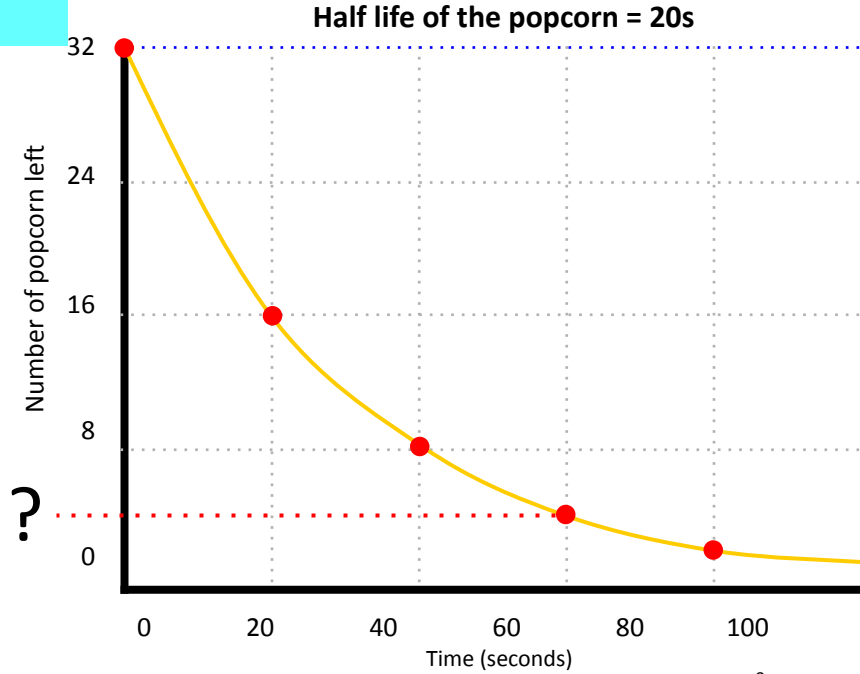


To know how different forms
of radiation behave.

The half life of the
popcorn is 20s

The sample
contains 32
popcorn at the
start.

How many will be
left after 60s?



Answer = 4

Question on half life

The half life of americium is 433 years.
The example contains 480 nuclei.

To know how different forms of radiation behave.

1. How many will be left after 1300 years?

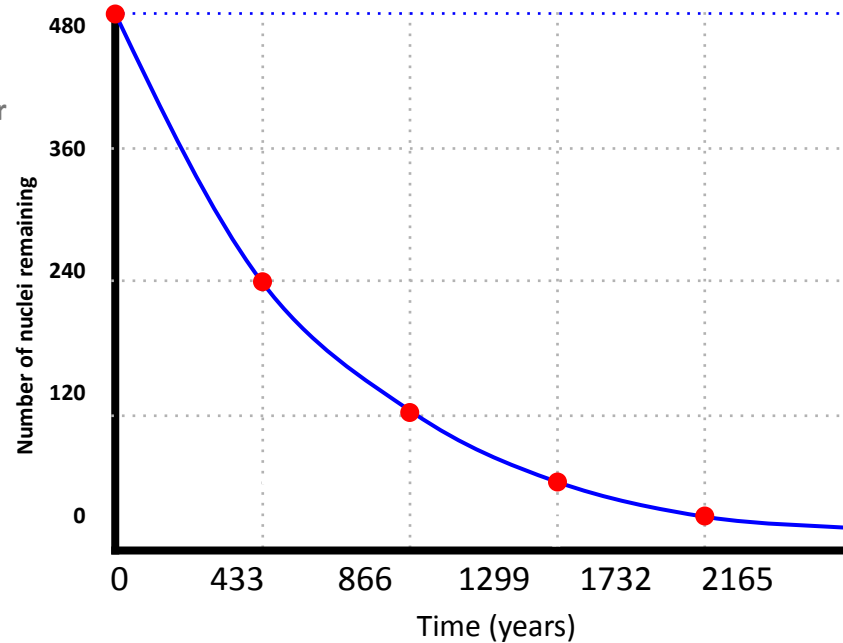
Answer1

2. How many years will have passed before it is harmless?

Answer 2

1. 60 nuclei

2. 6×433 years
= 2598 years

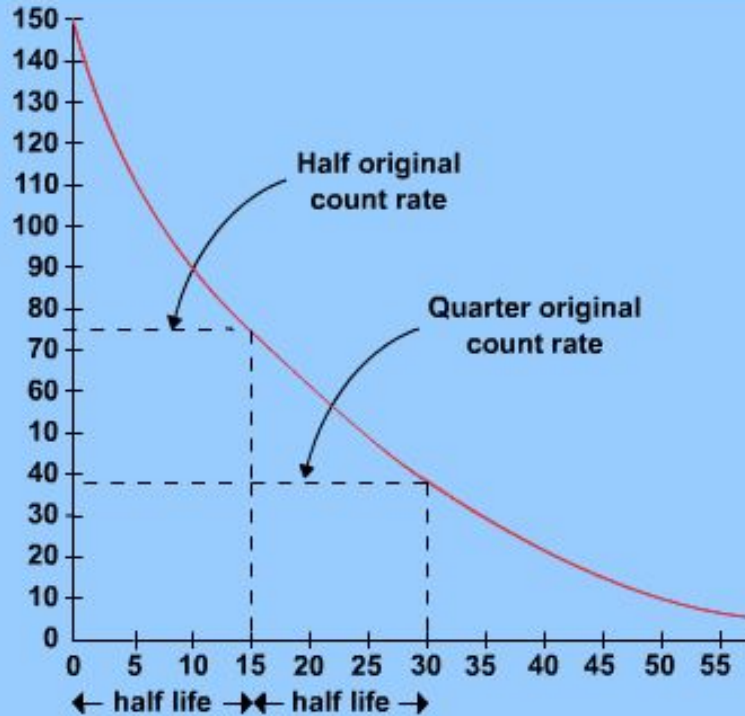


Explanation

Nuclei	480	240	120	60	30	16	8
Years	433	433	433	433	433	433	
Half Life	1	2	3	4	5	6	

Half-life:

Time/min.	0	5	10	15	20	25	30	35	40	45	50
Count rate/min.	150	118	92	75	63	52	38	35	26	19	17

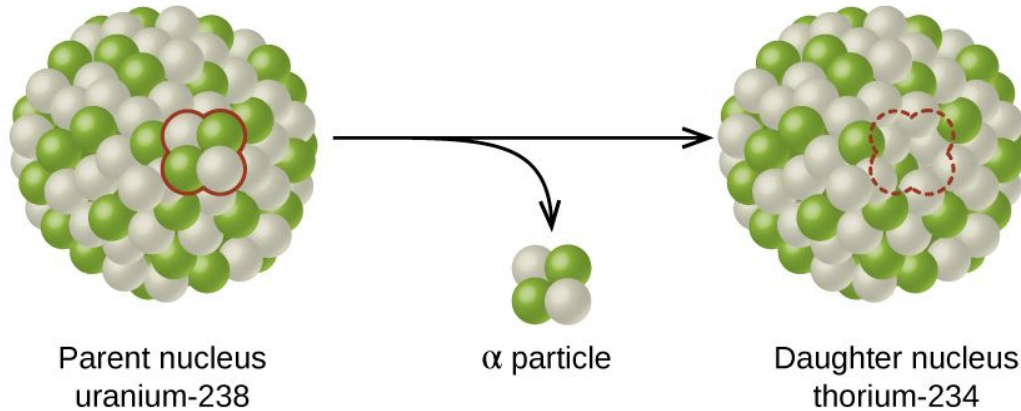


Half-life:

- Half life is the average time taken for half of the nuclei present to decay
- This is a prediction when the activity will fall to half its starting level

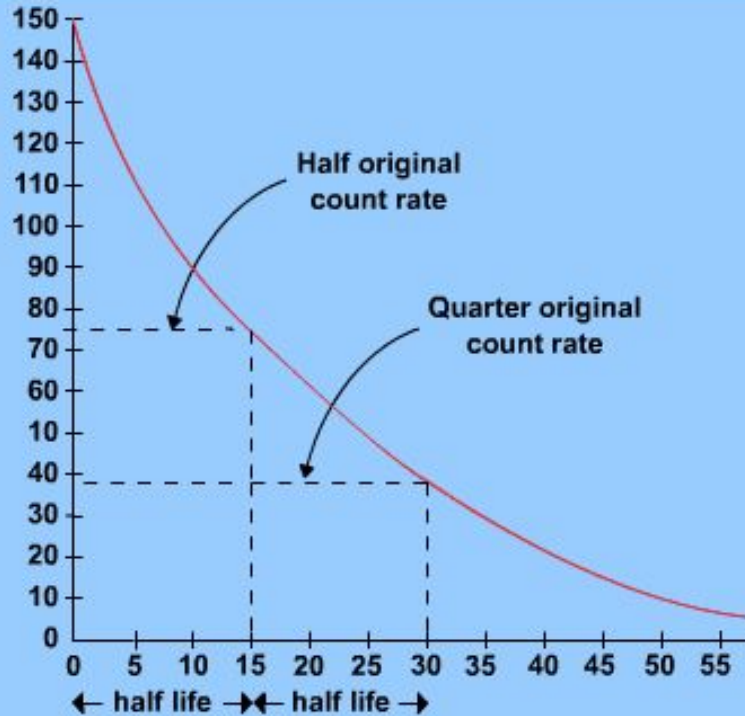
Half-life:

- In a sample you will have very many radioactive nuclei
- These will decay at random points
- Over time the activity of the sample will start to decrease as the nuclei decay and become stable



Half-life:

Time/min.	0	5	10	15	20	25	30	35	40	45	50
Count rate/min.	150	118	92	75	63	52	38	35	26	19	17



Half-life:

- Write a definition of half life
- The half life is the average time taken.....

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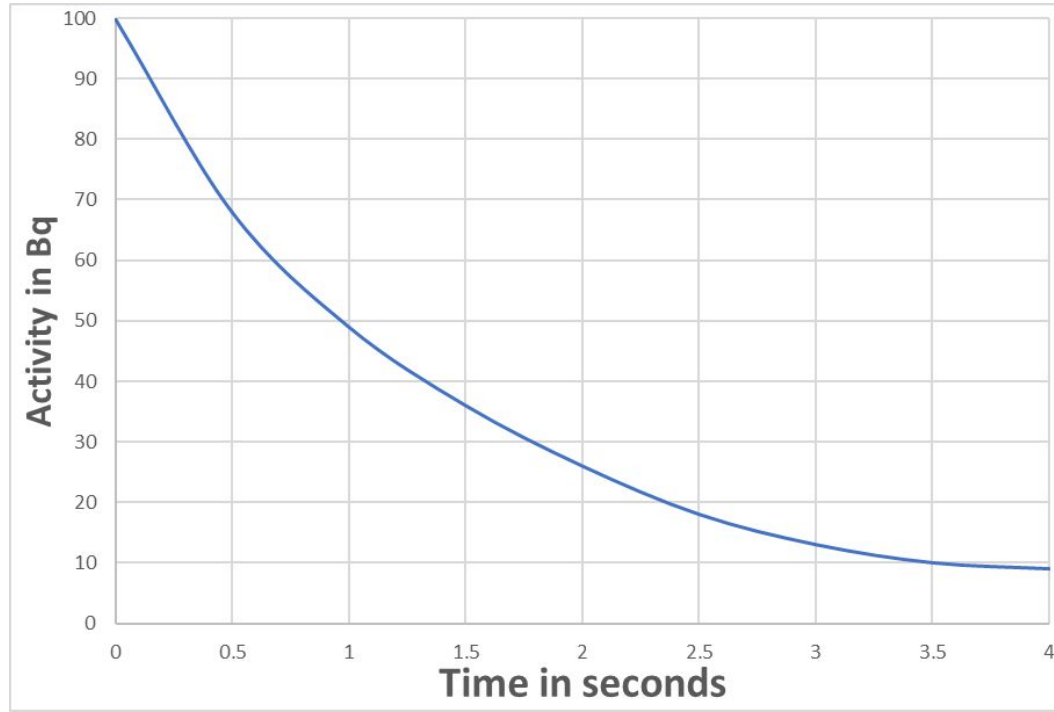
Calculating half-life:

- We can calculate the half-life using the graphs
- We must plot the data onto a graph

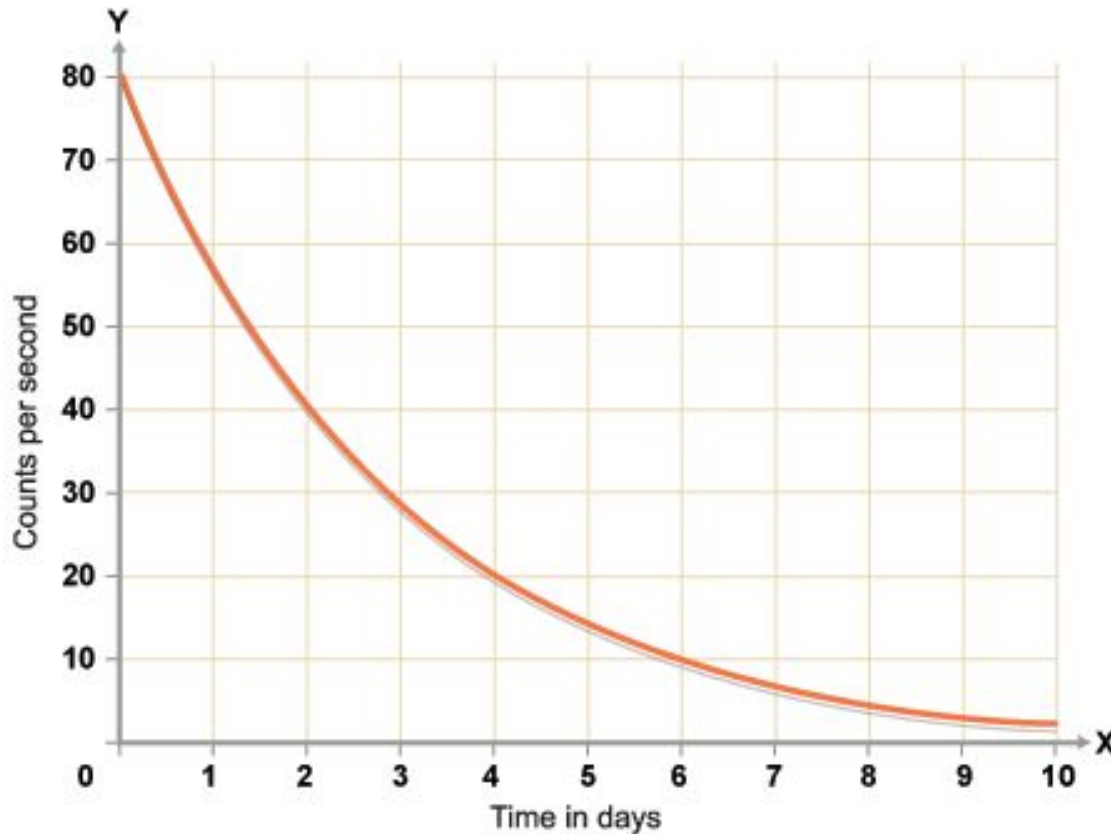
Time in minutes	Activity in Bq
0	100
0.5	76
1.0	51
1.5	40
2.0	26
2.5	18
3.0	12
3.5	10
4.0	8

Calculating half-life:

- We can calculate the half-life using the graphs
- We must plot the data onto a graph



Calculating half-life:



Calculating half-life:

- We may also work out the half life when given information of the decrease in activity and the time taken to decrease
- The activity of a radioactive sample took 4 hours to decrease from 100 Bq to 25 Bq. Calculate its half-life

Calculating half-life:

1. After 42 days the activity of a sample of phosphorus-32 has decreased from 400 Bq to 50 Bq. What is the half-life of phosphorus-32?
2. The half-life of radium-226 is 1600 years. If a sample of radium-226 has an original activity of 200 Bq, what will its activity be after 4800 years?

Calculating half-life:

Worksheet time!